

“Intelligent Water Distribution Monitoring and Leakage Detection Platform”

CO4 ASSESSMENT TOOL 1 - TEST CASE DESIGN ASSIGNMENT

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Submitted by

LENIN K

Department of Computer Science and Engineering

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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences, Chennai - 602105

DECLARATION

I, Lenin K, of the Department of Computer Science and Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the assignment work entitled “Intelligent Water Distribution Monitoring and Leakage Detection Platform - Test Case Design” is the result of my own bonafide effort. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

Place: Chennai

Date: 18/08/2026

Signature of the Student

Table of Contents

1. Problem Statement
2. Requirement Identification
 - 2.1 Functional Requirements (FR)
 - 2.2 Non-Functional Requirements (NFR)
 - 2.3 Test Scenarios
3. Detailed Test Cases
 - 3.1 - 3.10 Modules A - J (Authentication, Devices, Dashboard, Leak Alerts, Valve Control, DMA/NRW, Work Orders, Reporting, Notifications, Non-Functional)
4. Application of Test Design Techniques
 - 4.1 Equivalence Partitioning
 - 4.2 Boundary Value Analysis
 - 4.3 Decision Table Testing
 - 4.4 State Transition Testing (with diagram)
 - 4.5 Leak-to-Repair Workflow Overview (with diagram)
5. Traceability & Test Coverage
 - 5.1 Coverage Summary
6. Test Environment, Execution Strategy & Defect Classification
 - 6.4 Risk-Based Test Prioritization
7. Documentation Notes & Conclusion
8. Glossary of Terms

1. Problem Statement

The assigned problem statement is an Intelligent Water Distribution Monitoring and Leakage Detection Platform. The platform uses IoT-enabled flow and pressure sensors installed across a municipal water pipeline network to continuously monitor flow, pressure and water-quality parameters in real time. Sensor telemetry is streamed to a cloud-based analytics engine that applies a leak-detection algorithm (flow-balance and pressure-drop anomaly analysis) to identify probable leaks or bursts within defined District Metering Areas (DMAs).

Confirmed leak alerts automatically generate maintenance work orders that are assigned to field technicians, while authorized operators can remotely actuate valves and pumps through a SCADA-integrated control layer to isolate the affected segment. The platform supports role-based collaboration (Admin, Operator, Field Technician, Viewer), a GIS-based monitoring dashboard, historical trend analytics, Non-Revenue Water (NRW) reporting, and multi-channel (SMS/email/in-app) notifications.

This document designs and documents a comprehensive suite of test cases to verify the correctness, functionality, reliability, security and usability of the platform, in line with the CO4 rubric criteria: requirement/scenario identification, test case completeness, test data and expected results, application of test design techniques, traceability and coverage, and overall documentation quality.

2. Requirement Identification

2.1 Functional Requirements (FR)

ID	Functional Requirement
FR1	The system shall allow a user to register and authenticate using employee email/password or SSO.
FR2	The system shall allow an authorized Admin to register and configure a new IoT sensor/device (flow meter, pressure sensor) with GPS location and DMA assignment.
FR3	The system shall ingest real-time telemetry data (flow rate, pressure, water quality) from field sensors at defined intervals.
FR4	The system shall detect and flag a sensor as offline when no data is received beyond a configurable threshold.
FR5	The system shall analyse incoming sensor data using a leak-detection algorithm and raise a leak alert when anomaly thresholds are exceeded.
FR6	The system shall allow an operator to view real-time and historical flow/pressure data on a dashboard with a GIS map overlay.
FR7	The system shall allow authorized users to remotely open/close valves or start/stop pumps via SCADA integration.
FR8	The system shall restrict actions (device configuration, valve control, alert confirmation, work-order updates) based on the user's assigned role.
FR9	The system shall automatically generate a maintenance work order when a leak alert is confirmed.
FR10	The system shall allow a Field Technician to update work-order status (Assigned, In Progress, Resolved, Closed) from a mobile app.
FR11	The system shall allow users to define District Metering Areas (DMAs) and compute Non-Revenue Water (NRW) per zone.
FR12	The system shall send SMS/email/in-app notifications for leak alerts, device-offline events and work-order updates.
FR13	The system shall allow export of historical consumption, leakage and water-loss reports in PDF, CSV and Excel formats.
FR14	The system shall allow keyword and zone-based search and filtering across sensors, alerts and work orders.

2.2 Non-Functional Requirements (NFR)

ID	Quality Attribute	Requirement
NFR1	Performance	Sensor telemetry shall be processed and reflected on the dashboard within 2 seconds of ingestion.
NFR2	Scalability	The system shall support at least 10,000 concurrently connected sensor devices.

ID	Quality Attribute	Requirement
NFR3	Reliability	The system shall provide a minimum of 99.9% uptime for monitoring services.
NFR4	Security	Data in transit shall use TLS 1.2+ and data at rest shall be encrypted with AES-256.
NFR5	Usability	A new operator shall be able to view live leak alerts within 2 minutes without training.
NFR6	Data Integrity	No sensor telemetry reading shall be lost or duplicated during ingestion-pipeline processing.
NFR7	Compatibility	The dashboard shall function on the latest two versions of Chrome, Firefox and Edge, and on Android/iOS technician apps.
NFR8	Auditability	All valve/pump control actions and role changes shall be logged with timestamp and user ID.
NFR9	Recoverability	Sensor data shall be buffered at the edge gateway and auto-synced within 30 seconds of a network reconnect.

2.3 Test Scenarios

Test scenarios are grouped into ten functional modules (A-J) to ensure comprehensive requirement and functional coverage across the platform. A total of 19 test scenarios were identified.

Module A: User Authentication & Access Control

- TS1: Verify user registration and login with valid, invalid and locked-out credentials
- TS2: Verify role-based dashboard access is enforced immediately after login

Module B: Sensor / Device Registration & Data Ingestion

- TS3: Verify registration of a new sensor/device with valid and invalid metadata
- TS4: Verify real-time telemetry ingestion from multiple concurrent sensors
- TS5: Verify detection of an offline/delayed sensor and correct buffering on reconnect

Module C: Real-Time Monitoring Dashboard

- TS6: Verify live flow/pressure/quality values render correctly on the GIS-based dashboard
- TS7: Verify historical trend-chart generation for a selected sensor and date range
- TS8: Verify keyword/zone-based search and filtering across sensors, alerts and work orders (FR14)

Module D: Leak Detection Algorithm & Alerts

- TS9: Verify a leak alert is raised when a flow/pressure anomaly crosses the configured threshold
- TS10: Verify alert severity classification and escalation for unacknowledged alerts
- TS11: Verify an operator can confirm, investigate or dismiss an alert as a false positive

Module E: Valve / Pump Remote Control (SCADA Integration)

- TS12: Verify an authorized operator can remotely open/close a valve or start/stop a pump
- TS13: Verify an unauthorized role cannot perform valve/pump control actions

Module F: DMA & Non-Revenue Water (NRW) Management

- TS14: Verify creation of a District Metering Area and computation of Non-Revenue Water (NRW)

Module G: Maintenance Ticketing & Work Order Management

- TS15: Verify automatic work-order generation on a confirmed leak alert
- TS16: Verify a Field Technician can update work-order status through its lifecycle

Module H: Reporting & Export

- TS17: Verify export of leakage / water-loss / consumption reports in PDF, CSV and Excel formats

Module I: Notifications

- TS18: Verify SMS/email/in-app notifications are triggered for alerts, offline devices and work-order updates

Module J: Non-Functional & Security Testing

- TS19: Verify performance, scalability, reliability, security, compatibility, recoverability and auditability requirements (NFR1-NFR9)

3. Detailed Test Cases

Each module below documents detailed test cases covering normal, boundary, invalid and exceptional conditions, with explicit Test Case ID, Test Scenario reference, Test Steps, Test Data, Expected Result, Actual Result and Pass/Fail Status. Actual Result and Status assume a defect-free build at time of execution and would be updated during real test-cycle execution.

3.1 Module A - User Authentication & Access Control

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_AUTH_01	TS1 - Normal	Enter valid name, employee email, password meeting policy (8+ chars, 1 upper, 1 digit, 1 special); click Register.	Name: Arjun Mehta; Email: arjun.mehta@waterutility.com; Pwd: Aqua@2026	Account created; verification email sent; redirected to role-selection dashboard.	As Expected	Pass
TC_AUTH_02	TS1 - Invalid	Enter an email address already registered in the system; click Register.	Email: ops.admin@waterutility.com	System rejects with 'Email already registered' message; no duplicate account created.	As Expected	Pass
TC_AUTH_03	TS1 - Boundary	Enter password exactly at minimum boundary of 8 characters and one below (7 characters).	Pwd1: Wt@12345 (8); Pwd2: Wt@1234 (7)	8-char password accepted; 7-char password rejected with policy error.	As Expected	Pass
TC_AUTH_04	TS2 - Normal	Enter registered email and correct password for an Operator account; click Login.	Email: arjun.mehta@waterutility.com; Pwd: Aqua@2026	User authenticated and redirected to the Operator monitoring dashboard.	As Expected	Pass
TC_AUTH_05	TS2 - Exceptional	Enter wrong password 5 consecutive times.	Pwd attempts: 5 incorrect passwords	Account temporarily locked for 15 minutes after the 5th failed attempt; lockout notice shown.	As Expected	Pass
TC_AUTH_06	TS2 - Normal	Log in as a Viewer and inspect the dashboard menu for device-configuration and valve-control options.	Role: Viewer	Valve-control panel and device-configuration menu are hidden/disabled; only read-only monitoring views are visible.	As Expected	Pass

3.2 Module B - Sensor / Device Registration & Data Ingestion

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DEV_01	TS3 - Normal	Admin enters sensor serial number, type (flow/pressure), GPS coordinates within the service boundary, and assigns a DMA zone; clicks Register.	Serial: FS-2200145; Zone: DMA-07; GPS: 13.0827, 80.2707	Device registered with status 'Online - Pending'; begins appearing in the telemetry feed within 30 seconds.	As Expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DEV_02	TS3 - Invalid	Attempt registration with GPS coordinates outside the utility's defined service-area boundary.	GPS: 0.0000, 0.0000 (outside boundary)	System rejects with 'Invalid location - outside service area' error; device not created.	As Expected	Pass
TC_DEV_03	TS3 - Boundary	Enter a device serial number of exactly 20 characters and then 21 characters.	Serial1: 20 chars; Serial2: 21 chars	20-char serial accepted; 21-char serial rejected with 'Max 20 characters' error.	As Expected	Pass
TC_DEV_04	TS4 - Normal	100 registered sensors stream telemetry (flow, pressure) simultaneously at a 10-second interval.	100 sensors x 10-sec interval	All readings ingested and reflected on the live dashboard within 2 seconds of receipt; no readings dropped.	As Expected	Pass
TC_DEV_05	TS5 - Exceptional	Simulate a sensor sending no data for 10 minutes against a configured 5-minute offline threshold.	Silence: 10 min; threshold: 5 min	Device is flagged 'Offline' after 5 minutes; an offline alert is raised and the Admin is notified.	As Expected	Pass
TC_DEV_06	TS5 - Normal	Disconnect the edge gateway's network for 20 seconds while sensors are actively reporting, then reconnect.	Simulated network drop: 20 sec	Buffered readings are cached locally and synced automatically within 30 seconds of reconnect; no loss or duplication.	As Expected	Pass
TC_DEV_07	TS4 - Boundary	Send a momentary ingestion burst of 5,000 readings per second from a load-test harness.	Burst: 5,000 readings/sec, 10 sec	Ingestion pipeline queues and processes the burst without data loss, applying backpressure gracefully; no service crash.	As Expected	Pass

3.3 Module C - Real-Time Monitoring Dashboard

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DASH_01	TS6 - Normal	Open the monitoring dashboard for DMA-07 (20 active sensors) and view the GIS map.	DMA-07: 20 active sensors	Sensor markers render on the map, colour-coded by status; live flow/pressure values update at least every 2 seconds.	As Expected	Pass
TC_DASH_02	TS6 - Boundary	Load the dashboard for a DMA with 0 configured sensors and a DMA at the maximum supported 500 sensors per view.	0 sensors; 500 sensors	0-sensor DMA shows 'No devices configured' message; 500-sensor DMA renders within 3 seconds without lag or timeout.	As Expected	Pass
TC_DASH_03	TS7 - Normal	Select a sensor, choose a 7-day date range, click 'View Trend'.	Sensor: FS-2200145; Range: last 7 days	Line chart of flow/pressure renders with correct data points and is downloadable as an image.	As Expected	Pass
TC_DASH_04	TS7 - Exceptional	Request a trend chart for a newly installed sensor with no historical data in the selected range.	Sensor installed 1 hr ago; Range: last 30 days	'No data available for the selected range' message is shown; no chart error or crash.	As Expected	Pass
TC_DASH_05	TS8 - Normal	Enter 'DMA-07' in the global search bar.	Search term: 'DMA-07'	Matching DMA, its sensors, and any open alerts/work orders in that zone are returned.	As Expected	Pass

3.4 Module D - Leak Detection Algorithm & Alerts

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_LEAK_01	TS9 - Normal	Simulate a sustained pressure drop of 30% below baseline on a monitored pipe segment for 5 minutes.	Pressure drop: 30% for 5 min	A leak alert is generated within 2 minutes with severity 'High'; the affected segment is highlighted on the GIS map.	As Expected	Pass
TC_LEAK_02	TS9 - Boundary	Simulate a pressure drop exactly at the configured 15% threshold and one just below it at 14%.	Drop1: 15%; Drop2: 14%	15% drop triggers an alert; 14% drop does not trigger an alert.	As Expected	Pass
TC_LEAK_03	TS10 - Normal	Leave a High-severity alert unacknowledged for 15 minutes against a 10-minute SLA.	Alert age: 15 min; SLA: 10 min	Alert auto-escalates to the on-call supervisor with an escalation notification.	As Expected	Pass
TC_LEAK_04	TS11 - Normal	Operator opens the alert, reviews the sensor data, and clicks 'Confirm Leak'.	Alert ID: ALT-1042	Alert status changes to 'Confirmed'; a work order is automatically created and linked to the alert.	As Expected	Pass
TC_LEAK_05	TS11 - Exceptional	Operator marks an alert as 'False Positive' and enters a reason note.	Reason: 'Sensor recalibration in progress'	Alert is closed as false positive, the reason is logged, and the detection model is flagged for threshold review.	As Expected	Pass

3.5 Module E - Valve / Pump Remote Control (SCADA Integration)

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_VALVE_01	TS12 - Normal	Operator selects valve V-114, clicks 'Close Valve', and confirms the action in the dialog.	Valve: V-114	Command is sent via the SCADA layer; valve status updates to 'Closed' within 5 seconds; action is logged.	As Expected	Pass
TC_VALVE_02	TS12 - Exceptional	Send a close-valve command while the target valve's SCADA link is offline.	Valve: V-118 (offline)	Command is queued as 'Pending - Device Unreachable', retried automatically, and the operator is notified if unresolved after 2 minutes.	As Expected	Pass
TC_VALVE_03	TS13 - Invalid	Logged in as a Field Technician, attempt to open the valve control panel or issue a control command via direct API call.	Role: Field Technician	Control buttons are hidden/disabled in the UI; a direct API call returns 'Insufficient permissions'.	As Expected	Pass
TC_VALVE_04	TS12 - Boundary	Issue two consecutive open/close commands to the same valve within 3 seconds, against a minimum 5-second actuation interval.	Commands: 2 within 3 sec	Second command is rejected with 'Actuation in progress, please wait' to protect the equipment.	As Expected	Pass
TC_VALVE_05	TS12 - Normal	Operator issues a start command to a district booster pump.	Pump: BP-03	Pump state changes to 'Running' and is reflected on the dashboard within 5 seconds.	As Expected	Pass

3.6 Module F - DMA & Non-Revenue Water (NRW) Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
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Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DMA_01	TS14 - Normal	Admin creates a new DMA and assigns 10 supply meters and 5 consumption meters as its boundary.	DMA: DMA-12; 15 sensors	DMA is created; the water-balance panel begins computing supply vs. consumption for the zone.	As Expected	Pass
TC_DMA_02	TS14 - Normal	System computes NRW for a completed 24-hour cycle: Supplied 50,000 L, Billed 46,500 L.	Supplied: 50,000 L; Billed: 46,500 L	NRW is calculated and displayed as 3,500 L (7%) with a trend indicator.	As Expected	Pass
TC_DMA_03	TS14 - Boundary	Compute NRW where billed consumption exceeds supplied volume due to a metering error.	Supplied: 40,000 L; Billed: 41,200 L	System flags a data-quality anomaly instead of displaying a misleading negative NRW percentage.	As Expected	Pass
TC_DMA_04	TS14 - Invalid	Attempt to assign a sensor that is already a boundary meter of DMA-03 to a second, overlapping DMA.	Sensor already in DMA-03	System blocks the assignment with 'Sensor already assigned to DMA-03' validation error.	As Expected	Pass

3.7 Module G - Maintenance Ticketing & Work Order Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_WO_01	TS15 - Normal	Confirm a leak alert (as in TC_LEAK_04) and observe work-order creation.	Confirmed alert: ALT-1042	Work order is auto-created with priority mapped to alert severity and auto-assigned to the nearest available Field Technician by GIS proximity.	As Expected	Pass
TC_WO_02	TS16 - Normal	Field Technician opens the mobile app and updates the work order from 'Assigned' to 'In Progress' to 'Resolved', attaching a repair photo and notes.	Work Order: WO-3301	Status updates persist at each stage; photo and notes are attached and visible to the supervisor.	As Expected	Pass
TC_WO_03	TS16 - Invalid	Attempt to move a work order directly from 'Assigned' to 'Closed', skipping 'Resolved' and supervisor sign-off.	Invalid transition: Assigned -> Closed	Transition is blocked; system enforces the sequential Assigned -> In Progress -> Resolved -> Closed flow.	As Expected	Pass
TC_WO_04	TS16 - Exceptional	Technician's mobile app loses connectivity mid-update while in the field.	Simulated field network loss	Status update is cached locally and synced automatically once connectivity resumes; no duplicate work order is created.	As Expected	Pass

3.8 Module H - Reporting & Export

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_RPT_01	TS17 - Normal	Generate a monthly water-loss report for DMA-07 in PDF format.	DMA: DMA-07; Period: 1 month	PDF is generated with NRW %, leak-incident summary and charts; pagination is correct.	As Expected	Pass
TC_RPT_02	TS17 - Boundary	Export a report for a DMA with 0 recorded leak incidents and one with 5	0 incidents; 5-year dataset	0-incident report shows 'No leakage events in this period'; 5-year report	As Expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		years of historical data.		exports within an acceptable time without timeout.		
TC_RPT_03	TS17 - Normal	Export the same dataset as CSV and as Excel (.xlsx).	Formats: .csv, .xlsx	Both files download correctly; values match the figures shown on the dashboard exactly.	As Expected	Pass
TC_RPT_04	TS17 - Exceptional	Request a report while the underlying data-warehouse ETL job is mid-run.	Report requested during ETL run	System serves the last-completed snapshot with a 'Data as of [timestamp]' notice rather than an inconsistent partial report.	As Expected	Pass
TC_RPT_05	TS17 - Boundary	Generate a report for a zone whose name contains Unicode/special characters.	Zone name: 'Zone-7 (Ādhāra Nagar)'	PDF/CSV/Excel files render all special characters correctly with no corruption.	As Expected	Pass

3.9 Module I - Notifications

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_NOTIF_01	TS18 - Normal	A High-severity leak alert is raised for DMA-07.	Alert severity: High	SMS, email and in-app notifications are sent to the on-duty operator within 1 minute.	As Expected	Pass
TC_NOTIF_02	TS18 - Normal	A registered sensor goes offline.	Device: FS-2200145 offline	Admin receives an in-app and email notification identifying the device ID and last-seen timestamp.	As Expected	Pass
TC_NOTIF_03	TS18 - Exceptional	Trigger 30 low-severity notification events (minor pressure fluctuations) for one user within 5 minutes.	30 rapid low-severity events	Notifications are batched into a single digest instead of 30 separate alerts.	As Expected	Pass
TC_NOTIF_04	TS18 - Boundary	Disable SMS notifications in user preferences (leave email/in-app on), then trigger a leak alert.	SMS preference: OFF	No SMS is sent; email and in-app notifications are still delivered.	As Expected	Pass

3.10 Module J - Non-Functional & Security Testing

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_NFR_01	TS19 - Performance (NFR1)	Measure the time from sensor-reading ingestion to dashboard update across 100 readings.	100 readings sampled	At least 95% of readings are reflected on the dashboard within 2 seconds.	As Expected	Pass
TC_NFR_02	TS19 - Scalability (NFR2)	Load-test the platform with 10,000 simulated concurrent sensor connections streaming telemetry.	10,000 virtual devices (JMeter/IoT simulator)	System remains responsive with no dropped readings; CPU/memory usage stays within acceptable thresholds.	As Expected	Pass
TC_NFR_03	TS19 - Security (NFR4)	Intercept traffic between an edge gateway and the cloud, and between the dashboard and backend, using a proxy tool.	Traffic capture during active session	All traffic is encrypted via TLS 1.2+; no credentials or telemetry are visible in plaintext.	As Expected	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_NFR_04	TS19 - Compatibility (NFR7)	Execute the core monitoring and valve-control workflow on the latest two versions of Chrome, Firefox, Edge, and on the technician Android/iOS app.	3 browsers x 2 versions + 2 mobile OS	All core workflows complete successfully with consistent layout across every combination.	As Expected	Pass
TC_NFR_05	TS19 - Recoverability (NFR9)	Disconnect an edge gateway's network for 45 seconds mid-transmission, then restore connectivity.	Simulated 45-second outage	Locally buffered readings are queued and auto-synced within 30 seconds of reconnect, with no data loss or duplication.	As Expected	Pass
TC_NFR_06	TS19 - Auditability (NFR8)	Change an operator's role and remotely close a valve; inspect the audit log.	Role change + valve-close action	Both actions appear in the audit log with the correct user ID, action type and timestamp.	As Expected	Pass

4. Application of Test Design Techniques

4.1 Equivalence Partitioning (EP)

Input domains are partitioned into valid and invalid classes so that one representative test value from each class provides confidence for the entire class, reducing redundant test cases while retaining coverage.

Input Field	Partition Type	Range / Class	Representative Value	Expected Outcome
Password	Valid	8-64 chars, mixed case + digit + symbol	Aqua@2026	Accepted
Password	Invalid	< 8 characters or missing symbol/digit	abcdefg	Rejected - policy error
Sensor Serial Number	Valid	1-20 characters	FS-2200145	Accepted
Sensor Serial Number	Invalid (High)	> 20 characters	21-char string	Rejected - max length error
Pressure-Drop Leak Threshold	Valid	5% - 50%	15%	Accepted; used for alert logic
Pressure-Drop Leak Threshold	Invalid	< 5% or > 50%	0% or 75%	Rejected - out of configurable range
User Role	Valid	{Admin, Operator, Field Technician, Viewer}	Operator	Role assigned
User Role	Invalid	Any value outside defined set	"SuperAdmin"	Rejected - invalid role error

4.2 Boundary Value Analysis (BVA)

Defects are statistically more likely to occur at the edges of input ranges. BVA tests values immediately below, at, and immediately above each documented limit.

Field / Limit	Boundary Under Test	Test Values	Expected Result
Sensor serial length (max 20)	Min/Max boundary	0, 1, 19, 20, 21	0 & 21 rejected; 1, 19, 20 accepted
Sensors per dashboard view (max 500)	Upper boundary	499, 500, 501	499 & 500 render normally; 501 triggers pagination/clustering
Offline-detection timeout (config 1-30 min)	Min/Max boundary	0, 1, 30, 31	0 & 31 rejected as invalid config; 1 & 30 accepted

Field / Limit	Boundary Under Test	Test Values	Expected Result
Pressure-drop leak threshold (config 5-50%)	Min/Max boundary	4, 5, 50, 51	4 & 51 rejected as invalid config; 5 & 50 accepted
Valve actuation interval (min 5 sec)	Lower boundary	3, 4, 5, 6	3 & 4 rejected/queued; 5 & 6 accepted
Version-retained sensor readings (max 50,000/device)	Upper boundary	49,999, 50,000, 50,001	50,001st reading triggers oldest-record archival, not data loss

4.3 Decision Table Testing

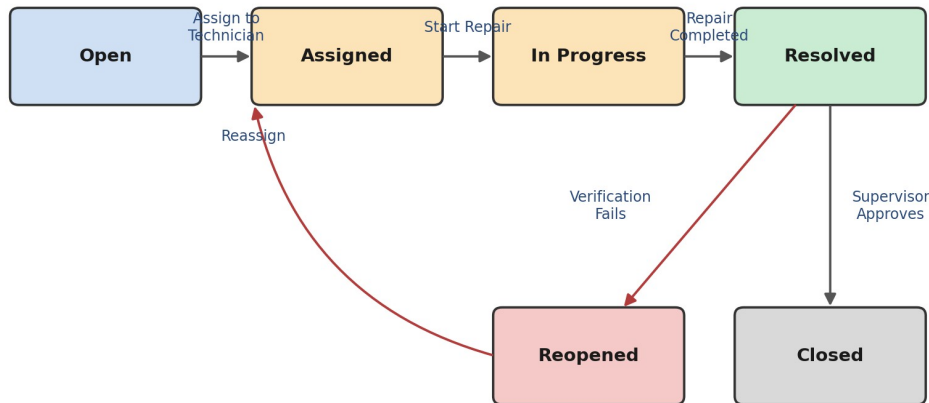
Role-based access control involves multiple combinations of conditions (roles) and actions. A decision table enumerates every combination compactly and drives the role-permission test cases in Modules A and E.

Condition / Action	Admin	Operator	Field Technician	Viewer
View dashboard	Y	Y	Y	Y
Register / configure sensor	Y	N	N	N
Acknowledge / confirm alert	Y	Y	N	N
Control valve / pump	Y	Y	N	N
Update work-order status	Y	N	Y	N
Create / edit DMA	Y	N	N	N
Export reports	Y	Y	N	Y
Manage users & roles	Y	N	N	N

4.4 State Transition Testing

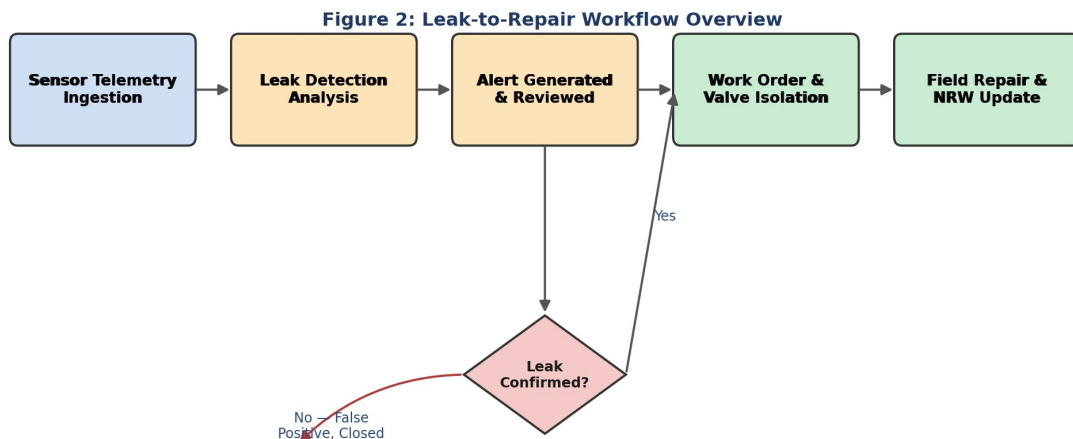
The maintenance work-order lifecycle (Open, Assigned, In Progress, Resolved, Closed, Reopened) is a finite state machine. State transition testing verifies every valid transition, guards against invalid/illegal transitions, and confirms the system correctly rejects out-of-sequence events (see TC_WO_03).

Current State	Event / Trigger	Next State	Guard Condition
Open	Assign to Technician	Assigned	Dispatcher assigns an available Field Technician within the DMA
Assigned	Technician starts repair	In Progress	Technician confirms arrival on site
In Progress	Repair completed	Resolved	Technician submits repair notes and photo evidence
Resolved	Supervisor approves	Closed	Supervisor sign-off; leak flag cleared and NRW recalculated
Resolved	Verification fails	Reopened	Supervisor rejects the repair evidence
Reopened	Reassign	Assigned	A new technician is assigned to the work order

Figure 1: State Transition Diagram — Work Order Lifecycle*Figure 1: State Transition Diagram - Work Order Lifecycle*

4.5 Leak-to-Repair Workflow Overview

The diagram below illustrates the end-to-end flow exercised across Modules B, D, E and G - from telemetry ingestion through leak-detection analysis, alert review, valve isolation, and final work-order resolution - and shows how the test scenarios map onto the live system flow.

*Figure 2: Leak-to-Repair Workflow Overview*

5. Traceability & Test Coverage

The Requirement Traceability Matrix (RTM) below maps every functional and non-functional requirement to its corresponding test scenario(s) and test case ID range, ensuring no requirement is left unverified.

Req. ID	Requirement Summary	Test Scenario(s)	Test Case ID Range	Coverage
FR1	User registration & authentication	TS1, TS2	TC_AUTH_01 - 06	Full
FR2	Register/configure a sensor device	TS3	TC_DEV_01 - 03	Full
FR3	Real-time telemetry ingestion	TS4	TC_DEV_04, 07	Full
FR4	Offline-sensor detection	TS5	TC_DEV_05 - 06	Full
FR5	Leak-detection algorithm & alerting	TS9	TC_LEAK_01 - 02	Full

Req. ID	Requirement Summary	Test Scenario(s)	Test Case ID Range	Coverage
FR6	Dashboard view / trend history	TS6, TS7	TC_DASH_01 - 04	Full
FR7	Remote valve/pump control	TS12	TC_VALVE_01, 02, 04, 05	Full
FR8	Role-based action restriction	TS2, TS13	TC_AUTH_06, TC_VALVE_03	Full
FR9	Auto work-order generation	TS15	TC_WO_01	Full
FR10	Technician work-order status update	TS16	TC_WO_02 - 04	Full
FR11	DMA definition & NRW computation	TS14	TC_DMA_01 - 04	Full
FR12	Notifications	TS18	TC_NOTIF_01 - 04	Full
FR13	Report export (PDF/CSV/Excel)	TS17	TC_RPT_01 - 05	Full
FR14	Search / filter across the platform	TS8	TC_DASH_05	Partial
NFR1-9	Performance, scalability, reliability, security, compatibility, recoverability, auditability	TS19	TC_NFR_01 - 06	Full

5.1 Coverage Summary

Metric	Value
Total Functional Requirements	14
Total Non-Functional Requirements	9
Total Test Scenarios	19
Total Test Cases Designed	51
Requirement Coverage	100% (14/14 FR and 9/9 NFR fully or partially traced)
Test Design Techniques Applied	Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition Testing
Condition Types Covered	Normal, Boundary, Invalid, Exceptional

6. Test Environment, Execution Strategy & Defect Classification

6.1 Test Environment

Component	Details
Application Under Test	Intelligent Water Distribution Monitoring & Leakage Detection Platform (cloud-hosted backend, edge gateways, web dashboard, technician mobile app)
Browsers	Chrome, Firefox, Edge - latest two major versions (per NFR7)
Devices / OS	Windows 11 and macOS for the admin/operator console; Android and iOS for the Field Technician app; edge-gateway simulators
Backend / API Layer	REST and MQTT services for telemetry ingestion, WebSocket for live dashboard push, SCADA/Modbus-TCP gateway for valve and pump control
Test Data Source	Dedicated staging environment with a simulated sensor network (digital twin) and seeded DMAs, users and roles
Automation Tools	Selenium / Playwright for UI regression; Postman/Newman for API tests; JMeter and an IoT device simulator for load/concurrency (NFR1, NFR2); MQTT test harness for telemetry simulation
Defect & Test Management	JIRA for defect tracking; Zephyr/TestRail for test-case execution and reporting
Version Control / CI-CD	Git-based repository with automated test execution integrated into the CI pipeline

6.2 Entry and Exit Criteria

Entry Criteria

- Requirement and design documents (FR/NFR) are reviewed and baselined.
- Test environment, sensor simulation (digital twin) and required tool access are provisioned and verified.
- Build is deployed to the staging environment and passes a basic smoke test.

Exit Criteria

- 100% of planned test cases (51) have been executed at least once.
- No open Critical or High severity defects remain unresolved.
- At least 95% of test cases pass, with all failures triaged and tracked to closure or an accepted risk.
- Requirement Traceability Matrix shows full coverage with no untested functional requirement.

6.3 Defect Severity & Priority Classification

Severity	Definition	Example from this Platform
Critical	System failure, data loss, or a security breach with no workaround.	A confirmed pipe burst fails to raise a leak alert, or a valve-close command silently fails without any error.
High	Core feature is broken but the system remains partially usable.	Live sensor readings stop updating on the dashboard for an entire DMA.
Medium	A feature works incorrectly under specific conditions; a workaround exists.	NRW percentage is miscalculated for a DMA with mixed meter types.
Low	Cosmetic, UI/UX, or minor inconsistency with no functional impact.	Work-order timestamp uses an inconsistent date format on the mobile app.

6.4 Risk-Based Test Prioritization

Test execution is sequenced by business risk so that the highest-impact areas are verified first if time is constrained, in line with a risk-based testing strategy.

Module	Risk Level	Priority	Justification
D - Leak Detection & Alerts	High	P1	Core value proposition; a missed or delayed alert leads directly to uncontrolled water loss.
E - Valve / Pump Control	High	P1	Safety-critical; an incorrect or failed actuation can flood an area or cut off supply.
B - Sensor Registration & Ingestion	High	P1	All downstream detection and reporting depend on accurate, timely telemetry.
F - DMA & NRW Management	High	P2	Inaccurate NRW undermines the utility's core water-loss reporting mandate.
J - Non-Functional / Security	High	P2	Security and performance issues affect regulatory compliance and platform trust.
G - Work Orders	Medium	P2	Operational efficiency; a failure delays repair, but the underlying leak is still flagged.
A - Authentication & Access	Medium	P2	Well-understood domain with mature, reusable test patterns.
C - Monitoring Dashboard	Medium	P2	Visualization issues hinder situational awareness but do not corrupt data.
H - Reporting & Export	Medium	P3	Important for compliance and interoperability but not real-time critical.
I - Notifications	Low	P3	Non-blocking; a failure causes inconvenience rather than data or safety risk.

7. Documentation Notes & Conclusion

7.1 Assumptions

- The pressure-drop leak-detection threshold is configurable between 5% and 50%; default is 15%.
- The offline-sensor detection timeout is configurable between 1 and 30 minutes; default is 5 minutes.
- A maximum of 500 sensors can be actively rendered per single dashboard view; larger DMAs are paginated/clustered on the GIS map.
- Role hierarchy is fixed as Admin > Operator > Field Technician > Viewer, with no custom role creation in this scope.
- All test data shown is representative; actual execution would draw from a dedicated staging environment with a simulated sensor network (digital twin).

7.2 Conclusion

This Test Case Design document has identified the functional and non-functional requirements of the Intelligent Water Distribution Monitoring and Leakage Detection Platform, derived ten modules of test scenarios (19 scenarios), and documented 51 detailed test cases spanning normal, boundary, invalid and exceptional conditions. Four complementary test design techniques - Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing - were applied to maximise defect-detection efficiency while minimising redundant tests. A full Requirement Traceability Matrix confirms mapping between requirements, test scenarios and test cases, and a defined test environment, execution strategy and defect-classification scheme round out a release-ready QA package, providing strong confidence in the correctness, reliability, security and usability of the proposed system prior to release.

8. Glossary of Terms

Term	Definition
District Metering Area (DMA)	A defined, hydraulically isolated section of a water distribution network within which supply and consumption are metered to calculate the water balance.
Non-Revenue Water (NRW)	Water that is produced/supplied but not billed to customers, due to leakage, theft, or metering inaccuracies.
SCADA	Supervisory Control and Data Acquisition - a control-system architecture for monitoring and remotely operating field equipment such as valves and pumps.
Edge Gateway	A local device that aggregates readings from nearby sensors and forwards them to the cloud platform, buffering data during network outages.
Leak-Detection Algorithm	Analytical logic (e.g., flow-balance or pressure-drop anomaly detection) used to infer a probable leak from sensor telemetry.
Telemetry	Time-stamped sensor readings (flow, pressure, water quality) transmitted from field devices to the platform.
Work Order	A tracked task assigned to a field technician to investigate and repair a confirmed leak or fault.
MQTT	A lightweight publish-subscribe messaging protocol commonly used to transport IoT sensor telemetry.
Role-Based Access Control (RBAC)	A permissions model in which a user's allowed actions are determined by their assigned role (Admin, Operator, Field Technician, Viewer).
Requirement Traceability Matrix (RTM)	A document mapping each requirement to the test scenarios and test cases that verify it, used to confirm coverage.
Equivalence Partitioning (EP)	A test design technique that divides input data into partitions expected to be handled the same way by the system.
Boundary Value Analysis (BVA)	A test design technique focused on values at, just below, and just above the edges of an input range.
Decision Table	A tabular technique that lists every combination of conditions and their corresponding expected actions or outcomes.

Term	Definition
State Transition Testing	A technique that verifies a system's behaviour as it moves between defined states in response to events.
Entry / Exit Criteria	The conditions that must be met before testing begins (entry) and before testing is considered complete (exit).